

Valid inequalities for polyhedra

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In this note we characterize the valid inequalities for a polyhedron. This characterization is a specialization of a more general result for general convex sets. We therefore present the general result first.

For a convex set $K \subseteq \mathbb{R}^n$, the set of directions along which one can start in K and move infinitely far within K is called the *recession cone* of K , and it is denoted $\text{rec}(K)$.

An inequality $\alpha^T x \leq \beta$ is called *valid* for a convex set $K \subseteq \mathbb{R}^n$ iff $\alpha^T x \leq \beta$ holds for all $x \in K$. If, in addition to validity, one has $\alpha^T \bar{x} = \beta$ for some $\bar{x} \in K$, then $\alpha^T x \leq \beta$ is called *supporting* for K .

The set of normal vectors of supporting inequalities for a convex cone $K \subseteq \mathbb{R}^n$ is called the *dual cone* of K , and the dual cone of K is denoted K° .

The *cone extension* of a convex set $K \subseteq \mathbb{R}^n$ is a convex cone $K^{\text{ext}} \subseteq \mathbb{R}^n \times \mathbb{R}_+$ that satisfies:

$$(i) \begin{pmatrix} x \\ 1 \end{pmatrix} \in K^{\text{ext}} \iff x \in K \quad \text{and} \quad (ii) \begin{pmatrix} x \\ 0 \end{pmatrix} \in K^{\text{ext}} \iff x \in \text{rec}(K)$$

The set of valid inequalities for a general convex set $K \subseteq \mathbb{R}^n$ can be described as follows (Theorem 3 in note 2.(D)):

Fact 1 (*Characterization of the valid inequalities for a convex set*)

Let $K \subseteq \mathbb{R}^n$ be a convex set, and let $\alpha^T x \leq \beta$ be an arbitrary inequality.

(i) If K is empty, then $\alpha^T x \leq \beta$ is valid for K .

(ii) If $\alpha^T x \leq 0$ is valid for $\text{rec}(K)$:

$$\alpha^T x \leq \beta \text{ is valid for } K \iff \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} \in (K^{\text{ext}})^\circ.$$

(iii) If $\alpha^T x \leq 0$ is not valid for $\text{rec}(K)$, and K is non-empty, then $\alpha^T x \leq \beta$ is not valid for K . ■

Fact 1 can be specialized to polyhedra as follows. Consider a polyhedron given as the solutions set for a finite number of inequalities:

$$P = \{x \in \mathbb{R}^n : Ax \leq b\}, \quad \text{where } A \in \mathbb{R}^{m \times n} \text{ and } b \in \mathbb{R}^m.$$

The cone extension of P is the polyhedral cone $C_p(A, b)$ given by:

$$\begin{aligned} C_p(A, b) &:= \left\{ \begin{pmatrix} x \\ \lambda \end{pmatrix} \in \mathbb{R}^{n+1} : Ax - \lambda \cdot b \leq 0_m \text{ and } \lambda \geq 0 \right\} \\ &= \left\{ \begin{pmatrix} x \\ \lambda \end{pmatrix} \in \mathbb{R}^{n+1} : \begin{pmatrix} A & -b \\ 0_n^T & -1 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix} \leq 0_{m+1} \right\}. \end{aligned}$$

For any polyhedral cone $C = \{y \in \mathbb{R}^n : Dy \leq 0_p\}$ with $D \in \mathbb{R}^{p \times n}$, the dual cone C° is given by $C^\circ = \{\alpha \in \mathbb{R}^n : \alpha = D^T u \text{ for some } u \geq 0_p\}$, i.e., C° is the set of conic combinations of the rows of D (proved in note 1.(E)). The dual cone $(C_p(A, b))^\circ$ of $C_p(A, b)$ is therefore given by:

$$\begin{aligned} &(C_p(A, b))^\circ \\ &= \left\{ \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} \in \mathbb{R}^{n+1} : \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} = \begin{pmatrix} A & -b \\ 0_n^T & -1 \end{pmatrix}^T \begin{pmatrix} u \\ u_0 \end{pmatrix} \quad \text{for some } \begin{pmatrix} u \\ u_0 \end{pmatrix} \geq 0_{m+1} \right\} \\ &= \left\{ \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} \in \mathbb{R}^{n+1} : \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} A^T u \\ b^T u + u_0 \end{pmatrix} \quad \text{for some } \begin{pmatrix} u \\ u_0 \end{pmatrix} \geq 0_{m+1} \right\} \\ &= \left\{ \begin{pmatrix} \alpha \\ -\beta \end{pmatrix} \in \mathbb{R}^{n+1} : \alpha = A^T u \text{ and } \beta \geq b^T u \quad \text{for some } u \geq 0_m \right\} \end{aligned}$$

where in the last equality we used that $\beta = b^T u + u_0$ holds for some $u_0 \geq 0$ if and only if $\beta \geq b^T u$. We can therefore formulate the following characterization of the valid inequalities for the polyhedron P :

Theorem 1 (*Characterization of the valid inequalities for P*)

Let $\alpha^T x \leq \beta$ be an arbitrary inequality.

- (i) If P is empty, then $\alpha^T x \leq \beta$ is valid for P .
- (ii) If $\alpha^T x \leq 0$ is valid for $\text{rec}(P)$:
 $\alpha^T x \leq \beta$ is valid for $P \iff \alpha = A^T u$ and $\beta \geq b^T u$ holds for some $u \geq 0_m$.
- (iii) If $\alpha^T x \leq 0$ is not valid for $\text{rec}(P)$ and P is non-empty, then $\alpha^T x \leq \beta$ is not valid for P . ■